

FIELD STUDY PREPARED FOR CRW ENGINEERING GROUP, INC.

# CRW's BD and PM staff hand-assemble every state, federal, and tribal proposal from scratch while the firm is running six simultaneous engineering openings, and a human-reviewed AI draft-and-checklist tool could return scarce PE and PM hours to billable work.

CRW Engineering Group runs on a steady stream of public-sector, tribal, and rural-infrastructure task orders, the kind of work that lives and dies on RFP and SOQ responses. At the same time, CRW's own careers page keeps an evergreen 'always hiring' posture and, as of this study, had six paid openings live at once across CAD, inspection, and three separate PE disciplines, at an 80-person firm. The people who would normally write the past-performance, staffing, and qualifications sections of a proposal are the same people the firm is short on. Nobody among CRW's three closest verified competitors (PND Engineers, R&M Consultants, DOWL) shows any public sign of AI-assisted business-development tooling either, so the field around CRW is as undifferentiated on this as CRW is.

Other

Anchorage and Palmer, Alaska

2026-07-19

Researched and drafted by Alaska AI's agent team, reviewed by a human before it reached you

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## WHAT WE SAW

# The homework

### THE BUSINESS

CRW Engineering Group is a full-service, Alaska-based civil engineering and land surveying firm with more than 80 staff across offices in Anchorage, Palmer, and Olympia, Washington. It offers one-stop planning, permitting, surveying, engineering, and construction-management services, spanning airport, civil, structural, electrical, mechanical, environmental, and geotechnical engineering, water and wastewater, transportation, and rural infrastructure. Its client base is heavily public-sector and tribal, municipalities, Alaska Native villages, and federal agencies.

### THE REAL PAIN

CRW is always on the lookout for exceptional engineers, surveyors, technicians, and support staff... Or contact us with your resume if you don't currently see an open position relevant to your skillset.

That evergreen careers-page posture sat alongside six simultaneously posted paid openings: CAD Technician, Electrical Engineer PE, Construction Inspector, Civil Engineer EIT Transportation, Civil Engineer PE Water/Wastewater, and Civil Engineer PE Transportation. For an 80-person firm, six open roles spanning CAD, inspection, and three separate PE disciplines at once points to a firm running short-handed against its project load broadly, not in one team.

Source: <https://www.crweng.com/careers/> and <https://www.indeed.com/cmp/Crw-Engineering-Group>

### THE FIELD AROUND THEM

- **PND Engineers, Inc.** none found  
[source](#)
- **R&M Consultants, Inc.** A GIS StoryMap for the Alaska Native Heritage Center, data visualization, not AI. No AI, chatbot, or automation otherwise found. [source](#)

### AI IN THIS INDUSTRY NOW

- Bechtel runs an internal AI tool queried over its own project documents, generating punch lists and pulling priorities from thousands of pages, a working production tool, not a pilot.  
[source](#)

- **DOWL** none found [source](#)

- 63 percent of ACEC member firms report having an AI strategy in place or actively being developed as of Q1 2025, an 11-point jump over the prior year. [source](#)

- Only 27 percent of surveyed AEC professionals report actually using AI in their operations, most firms with any AI activity are stuck between intent and real use. [source](#)

## THE OPPORTUNITY

**When a state, federal, or tribal RFP or SOQ solicitation drops, assemble a compliant, well-written response fast, pulling from CRW's own past project descriptions, resumes and qualifications, and boilerplate.**

This scored highest of the opportunities the research supported. It touches every discipline's pipeline in a firm whose revenue model is fee-for-service task orders, and none of CRW's three verified competitors show any public AI-assisted business-development tooling. It beats the more loudly-cited recruiting pain on fit, that pain is real but is a conventional ATS problem, not a genuine AI opportunity.

### WHAT THEY USE TODAY

Manual, template-based assembly from prior proposals and resume files, done by the same PE and PM staff the six open roles show are already stretched.

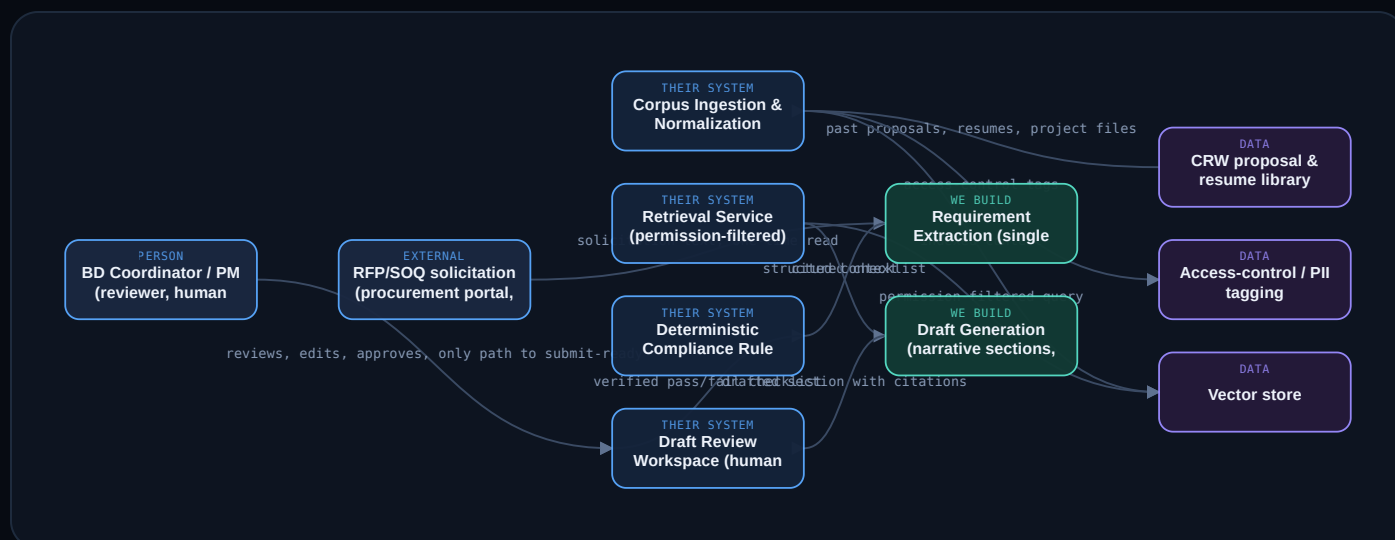
### WHY A CHANGE IS REALISTIC

The push is real, scarce PE and PM staff spend hours hand-assembling formulaic sections under public-sector deadline pressure. The pull is a first draft pulled from CRW's own past-performance library. The honest friction is that public solicitations carry strict compliance formatting, which raises real quality-anxiety about AI-drafted content, addressed below by keeping compliance checks deterministic rather than AI-judged.

## THE RECOMMENDED BUILD

### Proposal & SOQ Engine

A BD coordinator or PM feeds in a new RFP or SOQ. A single AI read of the solicitation extracts the objective compliance requirements, page limits, required forms, mandatory sections, into a checklist. A deterministic rule engine, not a second AI pass, verifies that checklist. Separately, a retrieval-plus-drafting step pulls from CRW's own ingested proposal library, resumes, and project descriptions to draft first-pass narrative sections, past performance, staffing, qualifications, each sentence traceable back to its source document. A human reviews, edits, and is the only one who can move a draft to submitted. No automated submission path exists or is planned.



■ Person ■ External ■ Their system ■ We build ■ Data

The solicitation enters on the left through a deterministic compliance gate. CRW's own proposal and resume library feeds a permission-aware retrieval layer that only the draft generator reads from. Every draft lands in a review workspace where a human, and only a human, can approve it.

#### WHERE AI FITS, AND WHERE IT DOES NOT

This is a workflow-tier build, not an autonomous agent, retrieval plus one drafting call, with a mandatory human gate before the one costly action. It earns that tier because matching a solicitation's specific requirement language against CRW's own past-performance content is a real retrieval-and-drafting problem a static template can't do as well. Where we

#### BUILD VS BUY

Buy the commodity, the embedding model, vector store, base language models, authentication, and observability tooling are all hosted services. Build the differentiator, the corpus ingestion pipeline that normalizes CRW's actual proposal and resume files, the permission-filtered retrieval layer, the deterministic compliance rule engine, and

would NOT use AI, the compliance checklist itself, page limits, required forms, mandatory sections, is an objective yes-or-no check. It runs as a deterministic rule pass after one AI read extracts the requirements, never as a second round of AI judgment. The same discipline ruled out two other candidates entirely, recruiting and applicant screening is a conventional ATS problem where a false negative costs a labor-scarce firm capacity it cannot spare, and a rural-client status portal rests on no sourced client complaint and would sit oddly against Alaska's relationship-based client culture. Neither needs AI, and neither is part of this build.

the review workspace that enforces the human gate.

## THE PLAN

### What we would build, scoped

CRW runs on a steady diet of state, federal, and tribal RFPs and SOQs. Every response needs past-performance write-ups, staffing bios, and qualifications narratives assembled from scratch, plus a compliance checklist gotten right. CRW is visibly running lean, six paid openings were live at once across CAD, inspection, and three PE disciplines at an 80-person firm, meaning the senior staff who write these sections are also the staff CRW is short on. This is inferred from CRW's revenue model and staffing pattern, not a proposal-specific complaint heard directly from CRW, and this study says so plainly.

#### GOALS

- Cut the hands-on time PM and BD staff spend hand-assembling narrative proposal sections.
- Catch objective compliance requirements before the human reviewer's first read, not after.
- Keep full human review and edit authority before submission intact.
- Stand up a permissioned, structured proposal and resume corpus CRW owns.

#### NON-GOALS, ON PURPOSE

- Any path to automated or AI-assisted submission of a proposal. (A wrong claim or missed compliance item can disqualify a bid or create false-claims exposure. A human reviews and submits every time.)
- AI judgment calls on whether a proposal is compliance-complete. (Objective yes/no checks run as a deterministic rule pass, never a second round of AI judgment.)
- Any drafting or review of stamped engineering content. (That work carries the sealing engineer's personal, non-delegable liability. This build never touches it.)
- Recruiting or applicant-screening automation. (A conventional ATS problem, not a genuine AI fit, evaluated and set aside.)
- A client-facing project-status portal for rural community clients. (Rests on

weaker evidence and, if built at all, should be a plain dashboard, not a conversational layer.)

## HOW WE WOULD MEASURE SUCCESS

| METRIC                                                                               | BASELINE                                                                  | TARGET                                                             | BY WHEN                                            |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------|
| PM and BD staff hours spent hand-assembling narrative proposal sections per response | Assumed, to be established with a 4-6 week time-log before Phase 1 launch | At least a 30 percent reduction versus the logged baseline         | First full quarter of proposal cycles after launch |
| Objective compliance misses caught only at final review or after submission          | Assumed zero formal tracking today                                        | Zero missed mandatory items past the deterministic checklist stage | First two full proposal cycles after launch        |
| Share of first-pass AI-drafted sections rated a usable starting point by reviewers   | Zero, no AI draft exists today                                            | At least 70 percent rated usable-as-a-starting-point               | First quarter of use                               |
| Pursuits submitted per quarter at the same staffing level                            | Unknown, to be pulled from CRW's own BD tracking                          | A measurable lift without added headcount                          | Two quarters following launch                      |

### PHASE ONE, MUST HAVE

- Ingest and permission CRW's existing proposal library, resumes, and project descriptions into a structured, access-controlled corpus.
- A single AI read of an incoming solicitation extracts objective compliance requirements into a checklist.
- A deterministic rule pass verifies that

### DELIBERATELY LATER

- An internal search assistant over CRW's project archive and regulatory library, riding the same ingestion pipeline.
- Extending drafting coverage to more solicitation types.
- A won or lost-bid feedback loop.



checklist.

- Retrieval-plus-drafting of first-pass narrative sections, tied back to source documents.
- A mandatory human review-and-edit gate before submission.
- A CRW-signed-off acceptance bar for what counts as a usable first draft.

WHAT WE WOULD NEED FROM YOU

- Access to CRW's existing proposal library and past submissions.
- Staff resumes and qualifications records, and a clear owner of resume PII permissioning.
- A named BD coordinator or PM as day-to-day point of contact.
- Four to six weeks of proposal-cycle hours logged for a real baseline.
- A recent sample of solicitation documents to validate the compliance-checklist logic.
- Sign-off that a human review-and-submit gate is the permanent operating model.

RISKS, AND HOW WE WOULD HANDLE THEM

| RISK                                                                          | MITIGATION                                                                                |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| The proposal-assembly pain is inferred, not a directly sourced CRW complaint. | Validate with a short baseline time-log before committing full Phase 1 budget.            |
| CRW's proposal library and resumes are likely scattered and inconsistent.     | Corpus ingestion is its own budgeted first phase, validated with a small pilot first.     |
| A wrong or missed compliance requirement can disqualify a bid.                | Mandatory human sign-off before submission, compliance checks run as deterministic rules. |
|                                                                               |                                                                                           |

## RISK

## MITIGATION

If reviewers find drafts unusable, the tool adds friction instead of removing it.

Define the acceptance bar with CRW staff up front, measure first-draft-usable rate from day one, halt further phases if it misses the bar.

## OPEN QUESTIONS WE WOULD RESOLVE TOGETHER

- What is CRW's actual PM and BD hours spent per proposal cycle today? (CRW BD lead, via a short time-log)
- How clean and centralized is CRW's existing proposal and resume library? (CRW IT or BD coordinator)
- What counts as a first draft good enough to route to a human for review? (CRW BD lead, defined jointly before the build starts)

## THE NUMBERS

# Honest ROI, not a hero number

### WHAT IT COSTS TO RUN, FIVE YEAR VIEW

The ask in this study is the Phase-1 pilot and nothing more: the corpus spike plus walking skeleton, a deterministic compliance checklist and drafted narrative sections run on one live pursuit using a small hand-loaded set of past proposals and resumes, priced at roughly \$20,000 to \$30,000. The pilot's case stands on its own terms. It is not priced to pay back in drafted hours, one pursuit's worth of saved drafting time is small, and we say so. What it buys is the decision data: a measured hours-per-proposal baseline where today there is none, a live test of whether the drafts are a real starting point for CRW's reviewers, and a go or no-go answer before any larger commitment. Weigh \$20,000 to \$30,000 of measurement against the \$176,000 to \$298,000 five-year build it gates, and against a field where most AI projects never reach the bottom line. Paying about a tenth of the build cost to find out whether the build is worth making is the conservative move, and it is the only spend this study asks for. The full build behind that gate, priced only as context: implementation and integration in year one (the biggest line, since CRW's proposal and resume corpus is unconfirmed-clean) an assumed \$65,000 to \$95,000, training and change management \$7,200 to \$18,000, ongoing run cost (inference, monitoring, evaluation, human review, corpus upkeep) \$18,000 to \$30,000 a year modeled over four and a half years for a mid-year-one go-live, and a 15 to 20 percent contingency on top. Computed five-year TCO: \$176,180 (lean) to \$235,705 (most likely) to \$297,600 (conservative, if the corpus is messier than assumed).

### WHERE THE VALUE COMES FROM

- **Senior PE and PM proposal-drafting hours saved per pursuit** CAPACITY Converts to cash by redeploying freed PE and PM time back onto billable stamped work, CRW is running six simultaneous openings, short-staffed rather than positioned to defer a hire.
- **BD coordinator compliance-assembly hours saved per pursuit** CAPACITY Converts to cash via redeploy of the coordinator's freed hours to additional pursuit prep.
- **Reduced risk of a compliance-technicality bid disqualification** CASH Booked as avoided cost, the smallest and least-certain of the three quantified lines. Modeled assuming 1 to 2 near-miss or lost pursuits a year at an assumed \$8,000 to \$15,000 in sunk pursuit cost each, with the checklist assumed to cut that risk 50 to 70 percent. No

CRW-specific disqualification history exists, this is a stated assumption, not a measured figure.

#### CONSERVATIVE

24 pursuits a year, 15 PE/PM hours and 6 BD hours per pursuit today, 25 and 50 percent time cuts, \$90 and \$45 loaded rates, one near-miss a year. Computed full run-rate benefit \$15,340 a year. With benefits at 30 percent of run-rate in year one (the stabilization ramp) and full run-rate in years two through five, cumulative five-year benefit is \$65,962 against the \$297,600 TCO, 22 percent recovered. The full build does not pay back within five years in this scenario.

#### MOST LIKELY

36 pursuits a year, 20 PE/PM and 8 BD hours per pursuit, 30 and 57.5 percent cuts, \$105 and \$52.50 rates, 1.5 near-misses a year. Computed full run-rate benefit \$41,724 a year. Same ramp, cumulative five-year benefit \$179,413 against the \$235,705 TCO, 76 percent recovered. The full build does not fully pay back inside the five-year window in this scenario either, breakeven trends past it.

#### AGGRESSIVE

48 pursuits a year, 25 PE/PM and 10 BD hours per pursuit, 35 and 65 percent cuts, \$120 and \$60 rates, 2 near-misses a year. Computed full run-rate benefit \$90,120 a year against the lean cost case. Breakeven lands at roughly month 24, and five years end at \$387,516 cumulative benefit against \$176,180 cumulative cost.

The conservative case still clears the bar. That is the case we would stand behind.

**Payback** For the pilot, the actual ask, the return is the decision data itself, a measured baseline and a go or no-go answer for about a tenth of the build cost, and that case holds in every scenario including the conservative one. For the full build the pilot gates, payback is roughly 24 months under aggressive assumptions only. The most-likely case recovers 76 percent of its five-year cost without fully paying back, and the conservative case recovers 22 percent. We computed these rather than narrating them, every figure above reproduces from the stated drivers and the stated ramp, and we are saying plainly that the full build only clears under aggressive assumptions. That is precisely why the only spend this study proposes is the pilot.

## The honest base rate

Most AI projects never move the bottom line, MIT puts it at roughly 95 percent, RAND at roughly 80 percent. This build is designed to sit in the smaller share that does, it is retrieval and one drafting call, not a chain of autonomous actions, it fits BD and PM staff's existing draft-then-review habit instead of requiring new behavior, and it keeps a human gate on the one costly action. Even with those guardrails, the full build's own numbers clear the conservative bar only at the pilot's ask size, which is exactly the honesty this method is built to surface rather than paper over. The pilot is the proposal. Further phases are funded only once the actual measured numbers, not the assumed ones, prove it out.

**Who owns the number** Brian Looney, President and the study's addressee, is the most plausible value owner and budget sponsor, since this study did not find a named BD director at CRW. The variance check, log actual PE, PM, and BD-coordinator hours per pursuit against the pilot's measured baseline for four full quarters after the pilot launches, and report actual versus modeled before funding any phase beyond it.

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## THE ROADMAP

# By confidence, not by calendar

### NOW

Ingest and access-permission CRW's own past proposals, resumes, and project descriptions, including a PII redaction pass.

Percent of the library ingested and access-controlled

Deterministic compliance-checklist extraction for one live RFP or SOQ.

Zero missed mandatory items on the pilot pursuit

First-pass narrative drafting for one live pursuit, with mandatory human review before anything is submitted.

PE/PM hours per proposal, baseline versus after, on the pilot pursuit

### NEXT

Expand corpus coverage and drafting to CRW's full range of recurring solicitation types.

Pursuits submitted per quarter

Build a won or lost-bid feedback loop.

Win rate on submitted pursuits, held or improved

### LATER

An internal search assistant over CRW's project archive and regulatory library, riding the same ingestion pipeline.

Time for an engineer to find a prior design solution or regulation

A first-pass regulatory and permitting checklist generator for rural water, sewer, and DOT projects.

PE hours per project spent on manual regulatory review

Next is earned only when Now's actuals on the pilot pursuit clear the bar, a measured reduction in hours, zero missed compliance items, and reviewers confirming the drafts were a real starting point. Later is earned only when Next's actuals clear its own bar. Two candidates the research turned up, recruiting automation and a rural client status portal, were evaluated and set aside rather than deferred, one is a conventional ATS problem and the other rests on no sourced client complaint, neither belongs on this roadmap.

## The honest part, what could make this not work

- The core proposal-assembly pain is inferred from CRW's revenue model and hiring pattern, not a complaint we heard directly from CRW. We are treating it as a hypothesis to validate with a short time-log, not a confirmed fact.
- The full build's ROI only clears the payback bar under aggressive assumptions. The conservative and most-likely cases do not fully pay back within five years on the numbers we modeled, and we computed those numbers rather than rounding them our way. That is why the only spend this study asks for is the pilot, whose return is the measured data that decides whether the build happens at all.
- CRW's real proposal and resume corpus has not been inspected. If it is messier or more scattered than assumed, ingestion could cost more and take longer than the ranges above.
- No CRW competitor shows public AI-assisted business-development tooling, but the absence of a public sign is not proof CRW would be first, only that nothing was found.

### WHAT THIS IS

## This is the free version of how we work

If this is useful, reply and we will walk through the assumptions and the pilot scope together. No pitch, no call required first. The full picture of how we work is at [alaskaaihq.com/services](https://alaskaaihq.com/services).

The full picture lives at [alaskaaihq.com/services](https://alaskaaihq.com/services).

## SOURCES

1. CRW offers start-to-finish, one-stop solutions for planning, permitting, surveying, engineering, and construction-management, more than 80 staff members, offices in Anchorage, Palmer, and Olympia WA <https://www.crweng.com/>
2. CRW's evergreen careers posture, always accepting resumes <https://www.crweng.com/careers/>
3. Six simultaneous paid openings across CAD, inspection, and three PE disciplines <https://www.indeed.com/cmp/Crw-Engineering-Group>
4. CRW leadership, Brian Looney (President), Shane Blanchard, Pete Bellezza, and the info@crweng.com contact <https://www.crweng.com/about/leadership/>
5. Competitor PND Engineers, Anchorage-headquartered civil/structural, arctic and waterfront projects, no AI found <https://www.pndengineers.com/>
6. Competitor R&M Consultants, Anchorage/Wasilla/Fairbanks/Juneau offices, GIS StoryMap for the Alaska Native Heritage Center, no AI found otherwise <https://www.rmconsult.com/>
7. Competitor DOWL, Juneau office, no AI found <https://www.dowl.com/>
8. Bechtel's internal AI tool over project documents, a production tool generating punch lists <https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/12/18/architecture-engineering-construction-sector-slow-to-adapt-ai-survey-shows>
9. 63 percent of ACEC member firms have or are developing an AI strategy as of Q1 2025, an 11-point increase year over year [https://www.acec.org/wp-content/uploads/2025/10/AI-Adoption-in-AEC-Summary\\_Primer\\_October\\_2025.pdf](https://www.acec.org/wp-content/uploads/2025/10/AI-Adoption-in-AEC-Summary_Primer_October_2025.pdf)
10. Only 27 percent of surveyed AEC professionals report actually using AI in their operations <https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/12/18/architecture-engineering-construction-sector-slow-to-adapt-ai-survey-shows>
11. MIT NANDA, The GenAI Divide: State of AI in Business 2025, about 95 percent of enterprise AI pilots deliver little to no measurable P&L impact <https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>
12. RAND, The Root Causes of Failure for Artificial Intelligence Projects, by some estimates more than 80 percent of AI projects fail <https://www.ailearnhub.org/resources/the-root-causes-of-failure-for-artificial-intelligence-projects-rand>

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